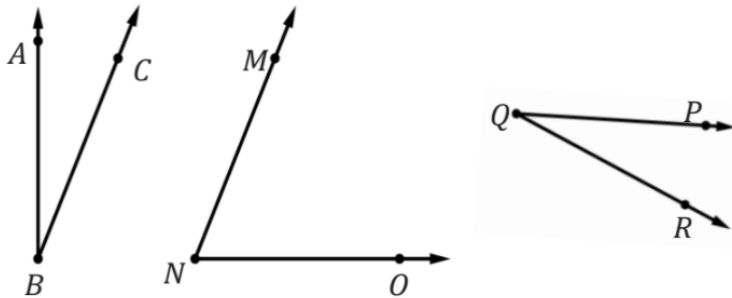




3.1.6 Wrap-Up: Cool Down

1. Three angles are shown with the given information.

$\angle ABC$ is complementary to $\angle MNO$
 $\angle MNO$ is complementary to $\angle PQR$
 $m\angle ABC = (5x - 3)^\circ$
 $m\angle MNO = 2(x + 22)^\circ$.



- a. Find the value of x .
- b. What is $m\angle PQR$?

Unit 3, Lesson 2: Vertical and Corresponding Angles



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



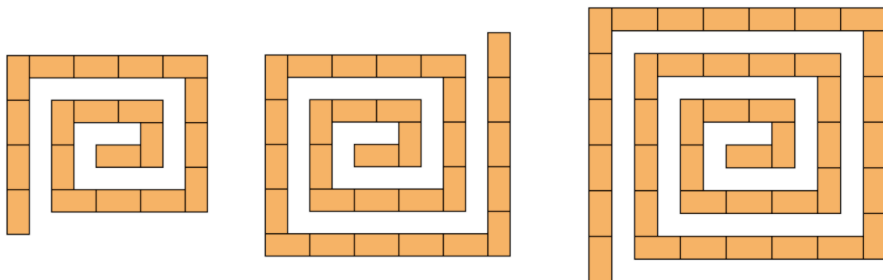
Learning Targets

- ☐ I can use vertical angles to solve problems.
- ☐ I can use corresponding angles formed by parallel lines and a transversal to solve problems.



3.2.1 Warm-Up: Shape Sequence

A sequence of three shapes created using rectangles is shown.

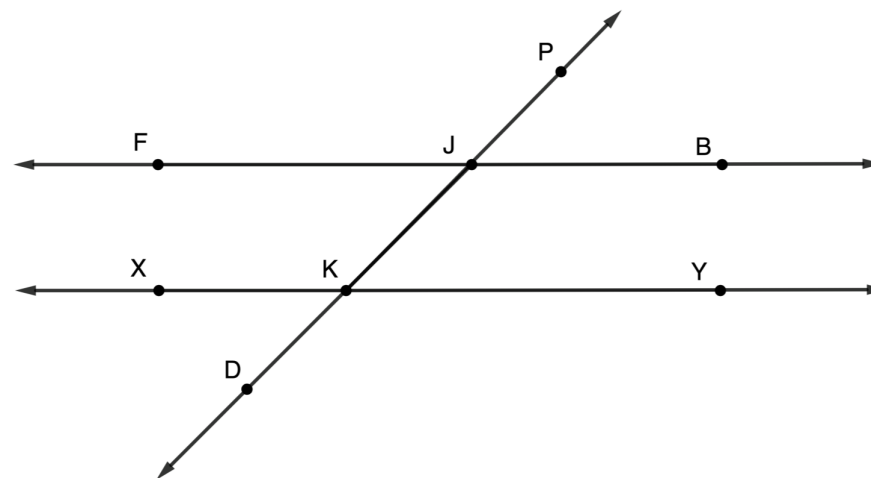


1. How many rectangles will have to be added for the next step in the sequence?
2. How many rectangles will be in Figure 11?



3.2.2 Exploration: Parallel Lines

$\overleftrightarrow{FB}$, $\overleftrightarrow{XY}$, and $\overleftrightarrow{DP}$ are shown, where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{FB}$ at point J and where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{XY}$ at point K .



1. Discuss with your partner what rigid transformations can be used to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K .
2. How can you use a translation to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K ?

In the diagram, $\overleftrightarrow{DP}$ is also known as a transversal, and $\overleftrightarrow{FB}$ and $\overleftrightarrow{XY}$ are parallel lines denoted as $\overleftrightarrow{FB} \parallel \overleftrightarrow{XY}$.



A **transversal** is a line that intersects two or more lines in the same plane at different points.

3. Use the translation of $\overrightarrow{FB}$ onto $\overrightarrow{XY}$ so that point J coincides with point K to name the new angles in the chart.

Pre-image	Image
$\angle PJF$	
	$\angle JKY$
$\angle DJB$	
	$\angle XKD$

The angles listed in the table above are corresponding angles.



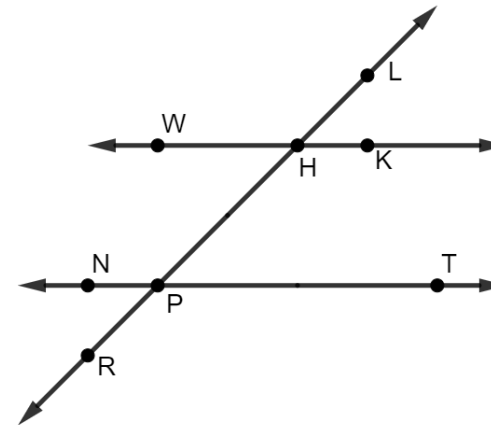
Corresponding angles are angles that are in the same position on two parallel lines in relation to a transversal.

4. Complete the statement.

Given that corresponding angles of parallel lines can be mapped to each other using rigid transformations, it can be concluded that the angles are

- ☐ complementary.
- ☐ congruent.
- ☐ supplementary.

5. In the figure shown, $\overrightarrow{WK} \parallel \overrightarrow{NT}$ where $\overrightarrow{PH}$ is a transversal.



- a. For each angle, name a corresponding angle and a vertical angle.

	$\angle TPR$	$\angle WHP$	$\angle KHP$	$\angle NPR$
Corresponding				
Vertical				

- b. Discuss your answers with your partner and make changes, if necessary.



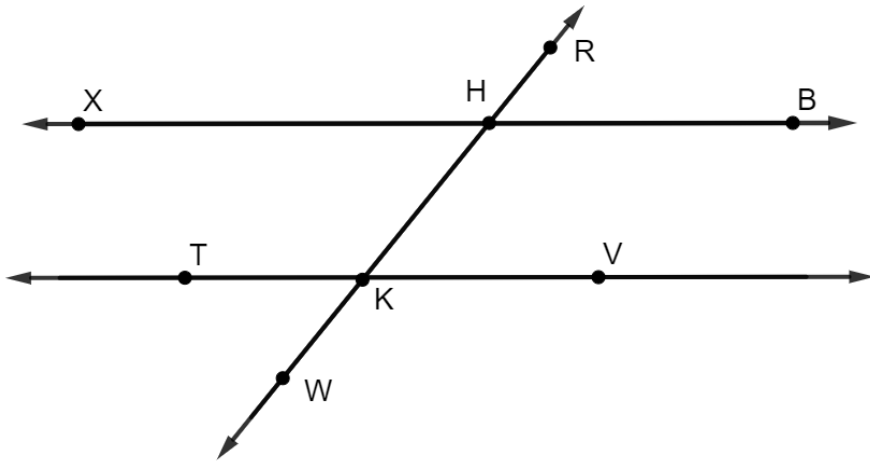
3.2.3 Guided Practice: Angles in Parallel Lines



Corresponding Angles Theorem

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown such that $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ is a transversal.



- a. Given $m\angle RHB = 51^\circ$, determine the measurement of each angle.

$$m\angle HKV =$$

$$m\angle TKW =$$

$$m\angle XHK =$$

- b. Describe how to use $m\angle RHB = 51^\circ$ to find $m\angle XHR$.
- c. Explain why knowing $m\angle XHR$ allows all of the angles to be determined.

2. $\overleftrightarrow{MN}$, $\overleftrightarrow{DK}$, and $\overleftrightarrow{PW}$ lie on a plane such that $\overleftrightarrow{MN} \parallel \overleftrightarrow{DK}$ and $\overleftrightarrow{PW}$ is a transversal. $\overleftrightarrow{PW}$ intersects $\overleftrightarrow{MN}$ at point Y and $\overleftrightarrow{DK}$ at point F . $m\angle PYN = (4x + 36)^\circ$, $m\angle NYF = (11y)^\circ$, and $m\angle KFW = 44^\circ$.

a. Draw a diagram that represents the given information.

b. Determine the value of y .

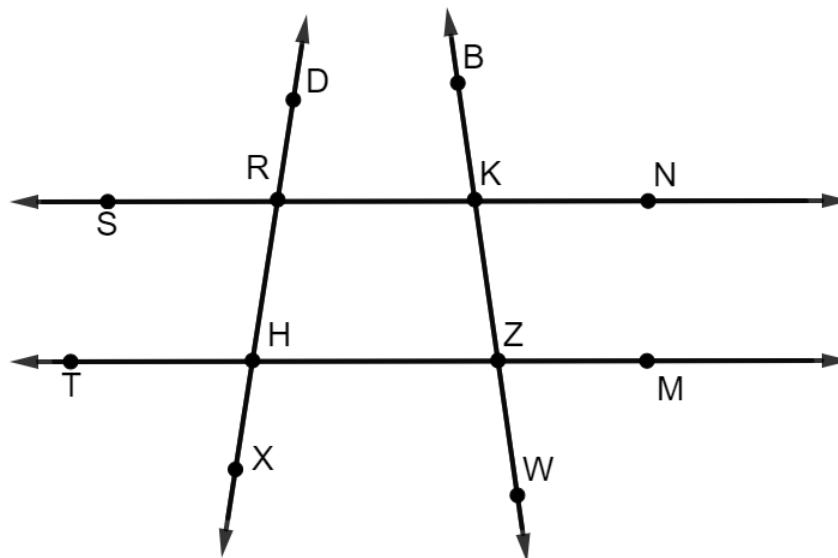
c. Determine the value of x .

d. What is $m\angle PYM$?



3.2.4 Your Turn!

1. $\overleftrightarrow{SN}$, $\overleftrightarrow{TM}$, $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ lie on a plane, such that $\overleftrightarrow{SN} \parallel \overleftrightarrow{TM}$ and $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ are transversals. $m\angle BKN = 99^\circ$ and $m\angle KRH = 99^\circ$.

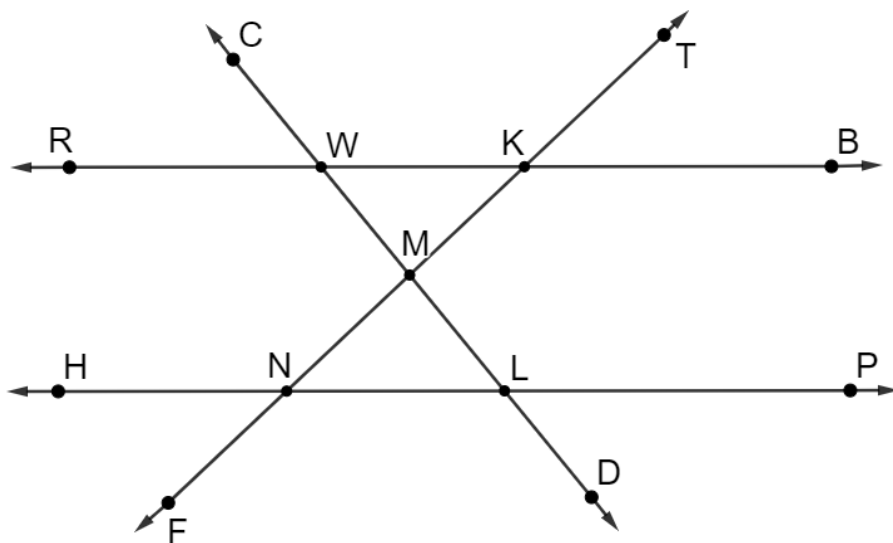


a. What is $m\angle KZM$?

b. What is $m\angle ZHX$?

c. What is $m\angle DRS$?

2. $\overleftrightarrow{CD}$, $\overleftrightarrow{TF}$, $\overleftrightarrow{RB}$, and $\overleftrightarrow{HP}$ are shown where $\overleftrightarrow{RB} \parallel \overleftrightarrow{HP}$ and $m\angle CWR = 51^\circ$ and $m\angle TKB = 43^\circ$.



Jasmyn's and Drew's justifications of the measure of $\angle CLH$ is shown.

Jasmyn's Answer	Drew's Answer
$m\angle CLH = 51^\circ$ because $\angle CLH$ and $\angle CWR$ are corresponding angles.	$m\angle CLH = 43^\circ$ because $\angle TKB$ and $\angle CWR$ are corresponding angles.

- a. Explain which student you agree with, including your reasoning.

- b. Find the measure of the following angles.

$$m\angle KNL = \underline{\hspace{2cm}}$$

$$m\angle HNF = \underline{\hspace{2cm}}$$

$$m\angle MLN = \underline{\hspace{2cm}}$$



1. Draw an example of vertical angles.
2. Draw an example of corresponding angles.

Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Target

- I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.